

Daniel Van Eijck

Senior Full-Stack Engineer · Backend focus · Rust · AWS

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PROFILE

Full-stack engineer with a backend focus. In my day job I've spent 6+ years building, securing and running multi-tenant IoT SaaS platforms on AWS for NZ and Australian enterprise customers in banking, aerospace, hospitality, cold-chain and aged care. That work spans device ingest pipelines, permission models, and the release and monitoring tooling that keeps them running, as well as the React and React Native apps on top, so I can take a feature from database to screen.

Outside my employment, I build independent projects on blockchain infrastructure, and that's where I write production Rust. The largest is a low-latency async trading engine (tokio, gRPC streams, websockets). It has an event-sourced core that can replay any production session deterministically, signing keys isolated at the crate level, and a test suite of around 2,000 tests that gates every release against recorded real-world data. I've also written CosmWasm smart contracts that hold real user funds, which have been through an external security audit by SCV Security and are covered by fuzz and mutation testing, plus Rust indexing pipelines feeding Postgres. Because mistakes there cost real money, it taught me to treat correctness, concurrency and key security as design constraints.

TECHNICAL SKILLS

Languages	Rust, Python, TypeScript, SQL; working knowledge of Solidity, Go, Java
Rust	tokio, tonic (gRPC), serde, tracing, CosmWasm, fuzz & mutation testing
AWS	Lambda, API Gateway, SQS, EventBridge, Cognito, Aurora/RDS, ECS Fargate, IoT Core, S3, CloudWatch, CDK, CodePipeline
Data	PostgreSQL, Redis, Hasura GraphQL, Elasticsearch, event sourcing
Security	Multi-tenant isolation, RBAC, JWT/Cognito, audit logging, OWASP ASVS, CI vulnerability scanning
Ops & delivery	Docker, Linux, GitHub Actions, Bitbucket Pipelines, AWS Amplify, Expo EAS, Traefik, nginx, Prometheus, Grafana, Loki, Sentry
Protocols	REST, GraphQL, gRPC, WebSockets, MQTT, CoAP, LoRaWAN
AI tooling	MCP servers, LLM agent integrations with role-gated, audited access, Claude Code
Frontend	React, TypeScript, Next.js, Vite, React Native / Expo, Apollo GraphQL, Tailwind, real-time charts & websockets

EXPERIENCE

Lead Web Services Software Engineer — Vention Lab

2023 – Present

Lead backend engineer across four production SaaS products for building and cold-chain monitoring, LPG monitoring, connected aged care and device ingest, all on AWS.

- **Multi-tenant authorisation.** Designed the policy-based access model for BenchMarker across an organisation → site → zone → asset hierarchy for ~70 customer organisations. Policies compile into a Postgres access table that Hasura row-level permissions enforce, so every query is scoped to the caller's organisation.
- **Device ingest hub.** Lead developer on Device Manager, which takes data from LoRaWAN network servers, AWS IoT and CoAP devices, normalises it into one format and routes it to downstream products. It handles ~720 gateways and ~1,400 active devices.
- **Serverless ingest (CDK).** Built an API Gateway → Lambda front door for device traffic. It buffers failed deliveries to SQS and replays them once the backend is healthy, with a DLQ and CloudWatch alarms, so an outage no longer loses customer data.
- **Security.** Led an OWASP ASVS Level 2 assessment of BenchMarker (260 controls, 54 findings) and drove the remediation, including a security summary for enterprise procurement. All customer data stays in Australian AWS regions. Built a CI security gate across 13 apps that blocks deploys with fixable high or critical vulnerabilities and keeps an audit trail for compliance evidence.
- **API access & audit logging.** Designed a role-based API used by all four backends, with hashed per-person keys, fail-closed confirmation for writes and a full audit log that includes denied attempts. Shipped a versioned, org-scoped Customer Data API with per-key rate limits.
- **CI/CD & operations.** Built a release register that links every deploy across 13 apps to its commit, security scan and rollback path. Fixed a deploy ordering bug that dropped jobs on schema releases, overhauled logging and monitoring (Prometheus, Grafana, Loki, Sentry), and found \$13–22k/yr in AWS savings.

- **Web apps.** Main contributor to BenchMarker's React web app. Led a new React/TypeScript building-management app that serves hotels, bank branches and offices from one codebase through per-customer configuration, including live HVAC control, occupancy and energy-savings reporting.
- **Mobile apps.** Build and ship the React Native/Expo apps for BenchMarker and GasOwl on iOS, Android and Web, with push notifications, in-app subscriptions and over-the-air updates through EAS.
- **Infrastructure.** Run self-hosted LoRaWAN network server and MQTT broker infrastructure, plus a gateway manager on EC2 with zero-touch provisioning and remote-access tunnels for field hardware. Services ship as Docker images through Bitbucket Pipelines, web apps through AWS Amplify, and mobile apps through EAS.
- **Team.** Mentor junior developers, review the team's pull requests, and write design docs and incident write-ups. Set up the shared development workspace and task tracker the team works from.

Software Engineer — Design Electronics Ltd (SenSys)

2020 – 2023

- **GasOwl serverless backend.** Built the backend largely solo: 26 Lambda functions behind API Gateway, Aurora PostgreSQL, Cognito, SQS and EventBridge, all defined in CDK and deployed through a multi-account pipeline with an approval gate before prod. Integrated Stripe and in-app subscriptions. It now runs 260+ installations across NZ and AU.
- **Ingestion & alarms.** Built BenchMarker's packet-processing pipeline (Celery/RabbitMQ), device decoders, and alarm notifications by SMS, email and push, routed by on-call rosters.
- Migrated REST APIs to GraphQL, added unit tests to the Django codebase, and built an Elasticsearch/Kibana analytics export.

Independent Rust & Distributed Systems Engineer — Choice Exchange / Injective ecosystem

2023 – Present

Production systems handling real user funds on the Injective blockchain, built alongside my full-time role.

- **Low-latency Rust trading engine** (tokio, gRPC, websockets), running in production. It uses an event-sourced design with deterministic replay, keeps signing isolated in one crate, and has ~2,000 tests. It evaluates ~250 routes in ~60 ms and wins ~80% of contested arbitrage opportunities. Rewriting it from Node.js to Rust cut latency variance from 200–500 ms to 50–150 ms.
- **Audited smart contracts (Rust/CosmWasm)** for Choice Exchange, a decentralised exchange: liquidity pools, vaults, staking rewards and an admin timelock. They have been externally audited by SCV Security, and are covered by fuzz and mutation testing, and have processed \$10.8M in cumulative trading volume, with ~\$600k in user deposits at the time of writing.
- **Indexing & API platform.** A Rust blockchain indexer writing to Postgres, feeding a Django/Celery/Redis/Hasura API behind a rate-limited gateway. It includes a live liquidity graph, cached in Redis, that powers a multi-hop order router.
- **Web frontends.** Built the Choice Exchange trading app and a React/TypeScript trading terminal that combines GraphQL, REST, websocket and gRPC-web feeds into real-time market views. Also built a token launch platform (Solidity contracts, indexer and real-time websocket API) with 200+ launches.
- **DevOps.** Everything runs in Docker on self-managed Linux servers. GitHub Actions handles testing, vulnerability scanning and container builds, and deploys to production; Prometheus, Grafana and Loki handle monitoring.
- **Security-critical services.** A transaction co-signing service, wallet-signature sign-in, and a crypto trading MCP server published on npm. The MCP server includes a policy engine (spend caps, allowlists), an encrypted keystore and an append-only audit log.

Research Assistant — Victoria University of Wellington

2019 – 2020

- Researched blockchain in supply chains: a report on using oracles to record each production step as an immutable, auditable record.

EDUCATION

Bachelor of Engineering with Honours — Victoria University of Wellington

2016 – 2020

OUTSIDE WORK

I train most days: gym and general fitness, plus tricking, an acrobatic sport that mixes martial arts kicks, flips and twists. Tricking has taught me a lot about breaking a hard skill into small, repeatable progressions and sticking with them, which carries over to engineering. I also have a background coaching gymnastics, where I learned to explain a skill clearly, adjust to how each person learns, and keep people safe while pushing them to progress.

REFERENCES

Available on request.